



Anatomical foundation models for brain MRIs[☆]

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ABSTRACT

Deep Learning (DL) in neuroimaging has become increasingly relevant for detecting neurological conditions and neurodegenerative disorders. One of the predominant biomarkers in neuroimaging is represented by brain age, which has been shown to be a good indicator for different conditions, such as Alzheimer's Disease. Using brain age for weakly supervised pre-training of DL models in transfer learning settings has also recently shown promising results, especially when dealing with data scarcity of different conditions. On the other hand, anatomical information of brain MRIs (e.g. cortical thickness) can provide important information for learning good representations that can be transferred to many downstream tasks. In this work, we propose AnatCL, an anatomical foundation model for structural brain MRIs that (i.) leverages anatomical information in a weakly contrastive learning approach, and (ii.) achieves state-of-the-art performances across many different downstream tasks. To validate our approach we consider 12 different downstream tasks for the diagnosis of different conditions such as Alzheimer's Disease, autism spectrum disorder, and schizophrenia. Furthermore, we also target the prediction of 10 different clinical assessment scores using structural MRI data. Our findings show that incorporating anatomical information during pre-training leads to more robust and generalizable representations. Pre-trained models can be found at: <https://github.com/EIDOSLAB/AnatCL>.

1. Introduction

Magnetic Resonance Imaging (MRI) is the foundation of neuroimaging, providing detailed views of the brain structure and function that are crucial for diagnosing and understanding various neurological conditions. However, the complexity and high dimensionality of MRI data present significant challenges for automated analysis. Traditional machine learning methods often require extensive manual annotation, which is both time-consuming and expensive. Recently, contrastive deep learning techniques have emerged as powerful tools for extracting meaningful information from complex data with minimal requirement for annotations, showing promise in improving the automated analysis of neuroimaging datasets. Contrastive learning methods leveraging weakly supervised loss functions have been shown to be effective in learning robust representation spaces [1,2]. These methods typically utilize metadata, such as the patient's age, to guide the learning process, to align similar data points (e.g., similar age) and distinguish

dissimilar ones in the learned representation space. While this approach has yielded promising results, it may fall short of fully capturing the multifaceted information inherent in MRI data and the anatomical complexity of the brain, as it primarily relies on age alone.

In this paper, we extend the capabilities of weakly supervised contrastive learning approaches by incorporating additional anatomical measures into the loss formulation. We propose AnatCL, a contrastive learning framework that improves upon previous works leveraging patient age as proxy metadata [1,3], to also include information derived from anatomical measures (such as cortical thickness), computed on region of interests (ROIs) of common atlas such as Desikan-Killiany [4] or Destrieux [5]. Specifically, inspired by the MIND anatomical similarity metric [6], we propose a modified contrastive learning loss function that integrates patient age with a targeted selection of three anatomical features: Mean Cortical Thickness, Gray Matter Volume, Surface Area. This selective integration was designed to assess the effectiveness of

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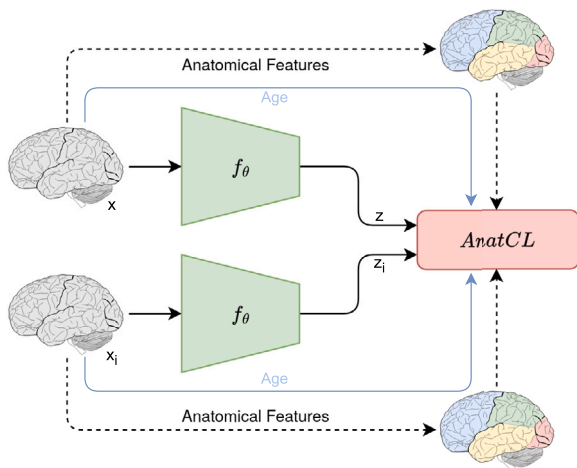


Fig. 1. We propose AnatCL, a contrastive learning approach that leverages patient's metadata (i.e., age) alongside anatomical information derived from structural MRIs, in order to learn meaningful and general representations. Given two input samples x and x_i , the objective of AnatCL is to learn embeddings z and z_i through a contrastive loss function, such that their distance in the representation space is proportional to the difference in age and brain anatomy between the two samples.

using potentially more informative anatomical features in conjunction with demographic data.

We propose and test two versions of our framework: a local version and a global version. The local version aims to capture the localized anatomical variations between specific ROIs (i.e., computing a cross-region similarity across all measures), while the global version aggregates these features across the entire brain to provide a more holistic view of the general anatomy (i.e., computing a cross-measure similarity across all regions). A graphical overview of AnatCL is provided in Fig. 1. We pre-train our models using the proposed AnatCL loss on the OpenBHB dataset, a large collection of healthy samples, and then we test the resulting models on different downstream tasks across a variety of publicly available datasets (Section 4.1). Our results show that in several downstream tasks, AnatCL achieves better results than traditional pre-training approaches based on brain age prediction only. These findings suggest that, in contrastive learning frameworks, incorporating anatomical features alongside the patient's age can enhance the expressiveness of the learned representations, leading to the development of models that learn more meaningful and generalizable representation spaces. Additionally, we also perform experiments to assess whether the embeddings learned using anatomical information can be leveraged for predicting clinical assessment scores, cognition, and psychopathology of the subjects (e.g., IQ, depression scores, and abnormal involuntary movements), which is typically performed on functional MRI data [7]. In summary:

1. We propose a novel weakly contrastive learning formulation that takes into account multiple features and metadata for guiding the learning process;
2. We propose AnatCL, a foundation model based on both anatomical and age features to build robust and generalizable representations of structural neuroimaging data;
3. We perform extensive validation across 12 different phenotyping tasks and the prediction of 10 different clinical assessment scores.

2. Related works

Many works have explored the potential of machine learning and deep learning in neuroimaging. Here, we review the most popular

approaches, distinguishing relevant topics for our work. First, we start with standard machine learning approaches, then we explore self-supervised methods, and finally, we present state-of-the-art contrastive approaches for neuroimaging.

Machine learning on structural brain imaging. Current literature has been mainly devoted to segmentation tasks for brain anatomical data, such as brain tumor segmentation [8] or tissue segmentation [9], in a supervised setting requiring annotated maps. The deep neural network U-Net [10] and its extension nnU-Net [11] were very successful at such tasks. Another body of literature tackled phenotype and clinical prediction tasks from structural imaging, such as age prediction [12], Alzheimer's disease detection [13], cognitive impairment detection, or psychiatric conditions prediction [14,15]. Current approaches mostly involve traditional machine learning models such as SVM or penalized linear models trained in a supervised fashion to predict the targeted phenotypes. Since most studies only include a few hundred subjects in their datasets, larger models introducing more parameters (such as neural networks) do not necessarily translate into better performance [2, 16,17], probably because of over-fitting.

Self-supervised learning on brain imaging. Self-supervised learning has been popularized by its impressive performance in Natural Language Processing (NLP) using auto-regressive models trained on web-scale text corpus and mainly deployed on generation tasks [18,19]. Since it does not require a supervised signal for training, it is particularly appealing when large-scale data are available, and the annotation cost is prohibitive. Its applicability on neuroimage data is still unclear and very few studies have tackled this problem so far [20]. Dufumier et al. [1] demonstrated that contrastive learning integrating subjects meta-data can improve the classification of patients with psychiatric conditions. Thomas et al. [21] demonstrated that learning frameworks from NLP, such as sequence-to-sequence autoencoding, causal language modeling, and masked language modeling, can improve mental state decoding from brain activity recorded with functional MRI.

Contrastive learning on brain imaging. Recently, interest in contrastive learning approaches for brain age prediction has risen. It has been demonstrated that contrastive learning performs better than traditional methods for neuroimaging [2]. Among different works, we can identify y-Aware [1] and ExpW [3]. These works propose to utilize contrastive learning for continuous regression tasks by employing a kernel to measure similarity between different samples. Ranking-based contrastive approaches [22] have also been proposed, showing promising performance on regression tasks. More recently, adaptive neighborhood selection strategies have been introduced in the context of brain age prediction [23,24], demonstrating improved generalization on datasets with non-uniform age distributions. In these works, age prediction is used as pre-training task, followed by transfer learning on specific diseases. However, they are limited to only considering age as pre-training feature. In this work, we extend this formulation to work with multiple attributes.

3. Method

In this section, we present the proposed method AnatCL. To introduce the adequate tools and concepts for understanding our work, we first provide an introduction to contrastive learning and weakly contrastive learning. Then, we present AnatCL.

3.1. Background

Contrastive learning (CL), either self-supervised as in SimCLR [25] or supervised as in SupCon [26], leverages discrete labels (i.e., categories) to define positive and negative samples. Starting from a sample x , called the *anchor*, and its latent representation $z = f(x)$, CL approaches aim at aligning the representations of all positive samples to

z , while repelling the representations of the negative ones. CL is thus not adapted for regression (i.e., continuous labels), as a hard boundary between positive and negative samples cannot be determined. Recently, there have been developments in the field of CL for tackling regression problems, especially aimed at brain age prediction, such as y-Aware [1] and ExpW [3], from brain MRIs. These approaches can be broadly categorized as weakly contrastive, as brain age is used as pre-train task.

3.2. Weakly contrastive learning with age

To tackle the issue of continuous labels, weakly contrastive approaches employ the notion of a degree of “positiveness” between samples [3]. This degree is usually defined by a kernel function $w_i = \mathcal{K}(y - y_i)$, where $0 \leq w_i \leq 1$ is computed by a Gaussian or a Radial Basis Function (RBF) kernel and y, y_i are the continuous attributes of interest (e.g. age). The goal of weakly contrastive learning is thus to learn a parametric function $f : \mathcal{X} \rightarrow \mathbb{S}^d$ that maps samples with a high degree of positiveness ($w_i \sim 1$) close in the latent space and samples with a low degree ($w_i \sim 0$) far away from each other. y-Aware [1] proposes to do so by aligning positive samples with a strength proportional to the degree, i.e.:

$$\mathcal{L}^{y\text{-aware}} = - \sum_i \frac{w_i}{\sum_j w_j} \log \left(\frac{\exp(\text{sim}(z, z_i))}{\sum_{t=1}^N \exp(\text{sim}(z, z_t))} \right) \quad (1)$$

where $z = f(x)$, $z_i = f(x_i)$, $z_t = f(x_t)$, and sim is a similarity function (e.g. cosine). A similar approach is also adopted by ExpW [3]. A limitation of these approaches is the reliance on one single attribute (age) to determine the alignment strength. They are thus not suited to leverage multiple attributes (e.g., anatomical measurements of the brain). Our proposed approach aims to solve this issue by extending the weakly contrastive formulation to include multiple attributes.

3.3. Anatomical contrastive learning

We propose a novel formulation for weakly contrastive learning to employ anatomical attributes derived from the MRI scan, which we call AnatCL. The features we consider are available in the OpenBHB dataset (see Section 4.1) for both Desikan-Killiany and Destrieux atlases, composed of 68 and 148 ROIs respectively. Inspired by MIND [6], we consider cortical thickness (CT), gray matter volume (GMV), and surface area (SA). Additional measures are available in the dataset; however, we did not achieve better results when using them (see supp. material). To include these measurements inside AnatCL, we propose two different formulations, global and local, based on how the similarity across two brain scans is computed.

Local descriptor. Given a sample x , we consider the set of local descriptors $\Psi(x) = \{\psi^1(x), \dots, \psi^K(x)\}$, where $\psi^k(x) \in \mathbb{R}^N$ for $k \in \{1, \dots, K\}$, K is the number of region of interest (ROIs) in the chosen atlas (68 for Desikan, 148 for Destrieux), and N is the number of features (i.e., 3). Each local descriptor is thus a vector composed of the three anatomical measurements CT, GMV, and SA, corresponding to a specific ROI of the atlas. To compute a degree of positiveness α_i based on the local descriptors of two samples x and x_i , we consider the average cross-region similarity as:

$$\alpha_i = \frac{1}{K} \sum_{k=1}^K s(\gamma(\psi^k), \gamma(\psi_i^k)) \quad (2)$$

where, for brevity $\psi^k = \psi^k(x)$ and $\psi_i^k = \psi^k(x_i)$, s can either be a similarity function or a kernel as in [1], and $\gamma(\psi)$ is a normalization function that normalizes each component of ψ to a standard range $[0, 1]$. In this work, for simplicity, we choose to employ cosine similarity as s . By plugging α_i into Eq. (1), we obtain the local variant of AnatCL (omitted the normalization terms $1/\sum_j \alpha_k$ and $1/K$ for brevity):

$$\begin{aligned} \mathcal{L}_{\text{AnatCL-L3}} = \\ = - \sum_i \sum_{k=1}^K s(\gamma(\psi^k), \gamma(\psi_i^k)) \log \left(\frac{\exp(\text{sim}(z, z_i))}{\sum_{t=1}^N \exp(\text{sim}(z, z_t))} \right) \end{aligned} \quad (3)$$

Global descriptor. Taking a complementary approach, we now consider global anatomy descriptors $\omega^j(x) \in \mathbb{R}^K$ for the entire brain with $j \in \{CT, GMV, SA\}$, that contain the values across all K regions for each anatomical measurement. Similarly to above, in order to compute a degree of positiveness β_i between two samples x and x_i given their global anatomical descriptors $\Omega(x) = \{\omega^{CT}(x), \omega^{GMV}(x), \omega^{SA}(x)\}$ and $\Omega(x_i) = \{\omega^{CT}(x_i), \omega^{GMV}(x_i), \omega^{SA}(x_i)\}$ we compute the cross-measurement similarity (global descriptor) as:

$$\beta_i = \frac{1}{3} \sum_j s(\omega^j, \omega_i^j) \quad j \in \{CT, GMV, SA\} \quad (4)$$

where, similarly to before, $\omega^j = \omega^j(x)$ and $\omega_i^j = \omega^j(x_i)$. This time we do not need to normalize ω^j and ω_i^j as s is evaluated between features of the same type, so the scale of the values is comparable. Also in this case, we employ a cosine similarity function for s . As before, by plugging β_i in Eq. (1) we obtain the AnatCL global loss (omitting the normalization terms):

$$\begin{aligned} \mathcal{L}_{\text{AnatCL-G3}} = \\ = - \sum_i \sum_j s(\omega^j, \omega_i^j) \log \left(\frac{\exp(\text{sim}(z, z_i))}{\sum_{t=1}^N \exp(\text{sim}(z, z_t))} \right) \end{aligned} \quad (5)$$

3.3.1. Final objective function

The final objective function is a combination of our proposed AnatCL loss (either local or global) and the y-Aware loss which considers age. Our final formulation is thus:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{AnatCL}} + \lambda_2 \mathcal{L}_{\text{age}} \quad (6)$$

where $\mathcal{L}_{\text{age}}$ is defined as in Eq. (1) and $\mathcal{L}_{\text{AnatCL}}$ is defined by either Eq. (3) or Eq. (5). The weights λ_1 and λ_2 control the contribution of each loss term independently. It is worth noting that the considered anatomical features in AnatCL can be computed directly from the MRI with standard tools such as FreeSurfer,² thus our method does not require any additional label in the dataset.

4. Experiments

In this section, we present the experimental data that we employ, and the experiments that we perform to assess the performance of AnatCL. We compare AnatCL to different baselines, composed of state-of-the-art approaches based on contrastive learning in several downstream tasks. An overview of downstream performance of AnatCL is provided in Fig. 2. From the plot, we can notice that AnatCL obtains state-of-the-art results in more tasks than any of the other methods.

4.1. Experimental data

We use a collection of T1-weighted MRI scans comprising 7,908 different individuals, for a total of 21,155 images. The experimental data is gathered from different publicly available datasets, and we target 5 different neurological conditions: healthy samples (OpenBHB [27]), Alzheimer’s Disease (ADNI [28] and OASIS-3 [29]), schizophrenia (SchizConnect [30]) and Autism Spectrum Disorder (ABIDE I [31]). Table 1 provides an overview of all the downstream tasks that we explore in this work.

OpenBHB. OpenBHB is a recently released dataset that aggregates healthy control (HC) samples from many public cohorts (ABIDE 1, ABIDE 2, CoRR, GSP, IXI, Localizer, MPI-Leipzig, NAR, NPC, RBP). Every scan comes from a different subject, for a total of 3984 MRI images. We use OpenBHB as the pre-training dataset for our method. Besides the structural scans and patients information, OpenBHB also provides 7 anatomical measures: cortical thickness mean and std, gray matter volume, surface area, integrated mean, gaussian curvature index, and intrinsic curvature index [27].

² <https://surfer.nmr.mgh.harvard.edu/>

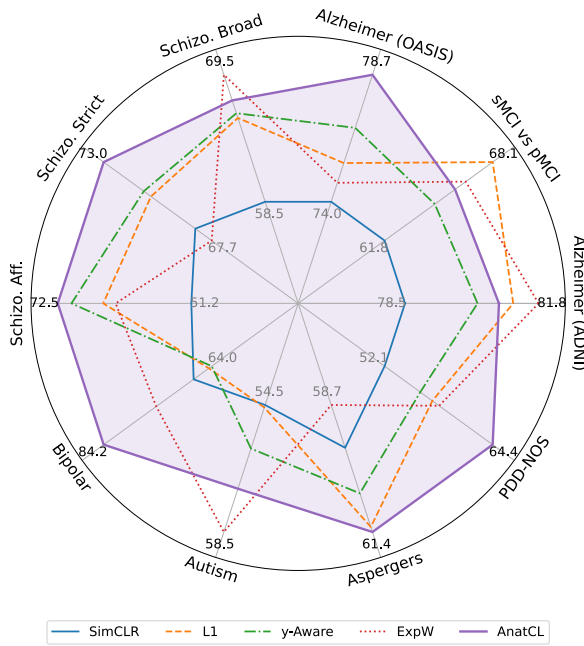


Fig. 2. Graphical evaluation of AnatCL (local and global) on all phenotyping tasks considered. Overall, AnatCL achieves higher results more consistently than any other baseline, with notable performance on ASD and Schizophrenia.

Table 1

Summary of downstream tasks (12) and clinical assessment scores (10) considered in the study.

Dataset	Task/Condition	Dataset	Phenotype
OpenBHB	Age (HC), Sex	SchizConnect	AIMS Overall Severity AIMS Upper Body AIMS Lower Body Depression Handedness SAS GAIT
ADNI	Alzheimer's Disease, sMCI vs pMCI	ABIDE 1	Handedness FIQ (WASI) VIQ (WASI) PIQ (WASI)
OASIS-3	Alzheimer's Disease		
SchizConnect	Schizo. Broad & Strict, Bipolar Disorder, Schizoaffective		
ABIDE 1	Autism, Aspergers, PDD-NOS		

ADNI and OASIS-3. The Alzheimer's Disease Neuroimaging Initiative (ADNI)³ is one of the largest and most popular datasets of MRIs for Alzheimer's Disease (AD). In our study, we include all phases ADNI-1, ADNI-2, ADNI-GO and ADNI-3 amounting to 633 HC, 712 mild cognitive impairment (MCI) and 409 AD patients, for a total of 14,385 MRI images. We downloaded all raw acquisition in the DICOM format and performed VBM preprocessing with CAT12 [32]. We also collected all clinical information and processed it in order to distinguish between stable and progressive MCI. Additionally, we also employ the OASIS3 dataset, including 685 patients (88 AD cases and 597 control cases), for a total of 1354 images.

SchizConnect. For schizophrenia, we employ SchizConnect, a large dataset of MRIs. We include structural MRIs for a total of 383 patients, divided into 180 HC, 102 with schizophrenia (broad), 74 with

schizophrenia (strict), 11 schizoaffective patients, and 9 with bipolar disorder, for a total of 383 MRIs.

ABIDE I. The Autism Brain Imaging Data Exchange (ABIDE) dataset is one of the most popular datasets of patients affected by conditions of Autism Spectrum Disorder (ASD). We leverage data from ABIDE-I, including structural MRIs for a total of 1102 patients, divided into 556 HC, 339 patients with autism, 93 patients with Asperger's Syndrome, and 7 with pervasive developmental disorder not otherwise specified (PDD-NOS). The total number of images is 1102.

Clinical assessment scores and phenotypes. In addition to predicting the diagnosis, we also consider some relevant clinical assessments included in the data, as reported in Table 1. For SchizConnect, we consider the Abnormal Involuntary Movement Scale (AIMS) evaluated in three assessments (overall, upper body, lower body), a Depression score based on the Calgary Scale, Handedness information, and GAIT measurements with the Simpson-Angus-Scale (SAS). For ABIDE, we also consider handedness and three IQ scores, namely Full Scale IQ (FIQ), Visual IQ (VIQ), and Performance IQ (PIQ) measured with the Wechsler Abbreviated Scale (WASI).

Data preparation and preprocessing. All the images underwent the same standard VBM preprocessing, using CAT12 [32], which includes non-linear registration to the MNI template and gray matter (GM) extraction. The final spatial resolution is 1.5 mm isotropic and the images are of size 121 x 145 x 121. The preprocessing is performed using the brainprep package.⁴ After preprocessing, we consider modulated GM images, as in [27].

Computation of anatomical features. Anatomical measurements on OpenBHB are released as part of the original dataset. For deriving the measurements, the brainprep module was employed, with FreeSurfer version FSL-6.0.5.1, and the Desikan-Killiany and Destrieux atlases as provided in FreeSurfer. We chose to employ CT, GMV, and SA as they are also used in [6]; for additional results on other measures, please refer to the supplementary materials.

4.2. Experimental setup

Our experiments involved pre-training a model using the proposed AnatCL loss on the OpenBHB dataset, followed by testing on various downstream tasks across different datasets. The experimental settings for the two loss formulations, local (AnatCL-G3) and global (AnatCL-L3), are identical. We pre-train two ResNet-18 3D models using VBM-preprocessed images and their corresponding anatomical measures with the proposed formulations. The training process employs the Adam optimizer with a learning rate of 0.0001 and a decay rate of 0.9 applied after every 10 epochs. The models are trained with a batch size of 32 for a total of 300 epochs. As values of λ_1 and λ_2 , for simplicity, we use 1. For the age loss (y-Aware) we use an RBF kernel with $\sigma = 2.0$, and the temperature parameter is set to 0.1. We do not employ any data augmentation, as positive pairs are defined based on age and/or anatomy labels. As standard practice in CL approaches [25,26], the contrastive loss is computed on a fully-connected projection head following the encoder, composed of two layers with dimensions 512 and 128. To ensure robust evaluation, we perform cross-validation using 5 folds. All data splits are at the subject-level to avoid leakage. The results are computed in terms of mean and standard deviation across the 5 folds. After the pre-training step we evaluate the models by testing their performance with a transfer learning approach: we extract the latent representations generated by the model using only the encoder of the model (i.e., discarding the fully connected head). For each downstream task, we train different linear classifiers on the

³ Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu).

⁴ <https://brainprep.readthedocs.io/en/latest/>

Table 2

Phenotyping results across healthy (brain age, sex), psychiatric (SCZ, ASD), and neuro-degenerative (AD) benchmarks in terms of mean absolute error (MAE) on age prediction, and balanced accuracy on all other tasks. AnatCL-L3 refers to the local version, while AnatCL-G3 refers to the global version; “3” refers to the three measures employed (CT, GVM, SA). *Indicates the some of the healthy subjects of ABIDE-I are included in OpenBHB. w/ aug indicates the use of data augmentation (random cropping and cutout).

Method	OpenBHB		Schizconnect				ABIDE-I*			ADNI		OASIS-3	Avg.
	Brain Age (I)	Sex	SCZ (Broad)	SCZ (Strict)	Schizoaff.	Bipolar	Autism	Aspergers	PDD-NOS	HC vs AD	sMCI vs pMCI	HC vs AD	
SimCLR	5.58±0.53	76.7±1.67	58.53±3.52	68.47±8.47	51.23±11.94	67.34±9.96	54.45±2.99	59.61±2.72	52.12±6.62	78.47±2.51	61.77±3.85	73.97±4.98	63.88
L1 (age-sup.)	2.73±0.14	76.7±0.67	65.79±4.74	70.68±5.10	65.24±15.21	64.49±22.08	54.53±1.79	61.33±8.77	57.54±5.93	81.20±2.30	68.12±5.42	75.40±5.40	67.37
y-Aware	2.66±0.06	79.6±1.13	66.20±4.50	71.04±2.31	70.29±14.73	63.95±19.81	55.84±3.37	60.60±9.22	56.17±10.17	80.30±1.80	64.72±4.43	76.70±3.30	67.76
ExpW	2.70±0.06	80.3±1.7	69.53±4.43	67.65±8.27	63.26±18.06	74.34±18.95	58.50±2.41	58.68±3.82	58.31±4.33	81.84±2.95	66.54±5.64	74.67±2.87	68.51
AnatCL-L3 (Desikan)	2.55±0.07	80.49±1.25	66.03±4.06	69.86±11.20	52.72±6.64	63.12±21.50	53.99±2.60	61.44±4.33	56.67±7.43	79.71±2.51	64.78±3.57	74.26±3.32	65.73
w/ aug	3.44±0.21	79.15±0.00	66.48±5.76	73.46±4.07	64.25±17.43	65.96±17.57	52.98±1.46	57.15±4.47	57.68±8.73	79.70±2.48	62.37±3.86	73.66±3.20	66.62
AnatCL-L3 (Destrieux)	2.58±0.08	80.0±1.0	67.31±5.75	70.03±7.16	72.47±12.27	55.88±13.23	53.14±4.20	60.99±4.94	56.63±6.74	79.38±3.56	64.97±3.40	77.00±2.28	67.07
w/ aug	3.41±0.10	80.95±0.00	64.53±11.54	70.94±6.15	68.14±12.36	62.74±17.35	55.60±0.80	59.06±4.92	57.17±6.36	79.09±1.77	63.45±5.30	70.16±4.18	66.53
AnatCL-G3 (Desikan)	2.56±0.06	79.0±1.0	64.31±4.43	72.46±4.41	62.42±14.95	72.42±15.77	57.12±2.79	58.25±4.46	64.39±9.26	80.84±2.38	64.88±4.37	78.32±4.48	68.58
w/ aug	3.44±0.15	79.68±0.52	62.47±8.17	74.25±6.52	59.69±22.02	64.77±16.73	57.89±2.09	57.33±5.01	55.36±5.72	77.27±2.59	63.41±3.81	71.41±3.17	65.78
AnatCL-G3 (Destrieux)	2.59±0.05	80.0±1.0	64.12±3.18	73.04±4.42	59.76±15.20	84.19±7.10	55.41±4.55	58.96±7.52	58.10±4.63	80.75±3.35	65.91±2.23	78.66±2.89	70.0
w/ aug	3.44±0.10	79.30±0.73	65.26±6.66	74.31±6.11	64.97±16.37	61.84±18.26	54.18±2.98	57.34±5.92	60.63±10.72	80.89±2.45	64.50±3.01	70.91±3.81	66.74

extracted representations to assess the model’s ability to learn meaningful and generalizable features. We run our experiments on the Jean Zay and Leonardo clusters (NVIDIA V100 and A100 GPUs), with a single training taking around 10 h.

Baselines for comparison. In this work we focus on contrastive learning as it has been demonstrated to work better than non-contrastive method in recent literature [2]. For this reason, we pick recent state-of-the-art contrastive methods in neuroimaging, namely y-Aware [1] and ExpW [3]. In addition to these recent techniques, we also consider the popular SimCLR framework [25] as a self-supervised contrastive baseline, also employed by BrainIAC [33]. For these experiments, we employed random volume cutout ($32 \times 32 \times 32$ voxels) and random cropping of the 3D volume. For comparison with traditional non-contrastive supervised techniques, we also include baseline models trained with L1 loss for brain age prediction [34].

4.3. Results

We evaluate the results of AnatCL pre-trained on OpenBHB by linear probing on different datasets, for several phenotyping and clinical assessment scores prediction tasks.

4.3.1. Brain age and sex prediction

The preliminary downstream tasks we evaluate are brain age prediction and sex classification, on OpenBHB. The results are reported in Table 2. AnatCL achieves state-of-the-art results on brain age prediction, reaching the lowest absolute error of 2.55 years. It is worth noting that we do not employ any bias-correction method [35], but we consider the raw predictions of the model. For sex classification, AnatCL is competitive with ExpW [3], and improves over the other baselines.

4.3.2. Schizophrenia and ASD

We evaluate phenotyping performance on SchizConnect for detecting schizophrenia (broad and strict), schizoaffective, and bipolar patients. Results are reported in Table 2. With AnatCL, we achieve state-of-the-art performance on three out of four tasks, suggesting that including anatomical information during training can prove useful for these psychiatric conditions. We also assess detection performance for Autism Spectrum Disorder (ASD) patients, across three categories (autism, aspergers, and PDD-NOS). While AnatCL does not beat the other methods on general autism (but is competitive), we achieve the best results on Aspergers and significantly improve the accuracy for PDD-NOS patients, which is a rarer diagnosis in the data. Overall, AnatCL achieves better results than other baselines most of the time, and is competitive with SOTA results in the remaining instances.

4.3.3. Alzheimer’s disease and cognitive impairments

In Table 2 we also report results for Alzheimer’s Disease (AD) detection on both ADNI and OASIS-3. On ADNI, we also assess the discriminative power between stable MCI (sMCI) and progressive MCI (pMCI) patients. While we do not reach the best results overall on ADNI, AnatCL can consistently improve over the self-supervised baseline SimCLR and over the weakly supervised method y-Aware, and reaches overall competitive results. On OASIS-3, AnatCL also achieves the best results in AD detection, in both local and global variants. Overall, AnatCL provides more consistent performance than other methods.

4.3.4. Clinical assessments scores

As final experiments, we turn our attention to predicting clinical assessment scores, cognition, and psychopathology from brain MRIs. This could provide useful insights on the relation between brain anatomy and individual phenotypes, whereas most of the literature focuses on fMRI [7]. The 10 scores considered can be distinguished based on the prediction task: AIMS, depression, handedness, and GAIT are classification tasks, while IQ scores (FIQ, VIQ, and PIQ) are regression tasks. For handedness, we predict right-handed vs other (left-handed or ambi), for depression, we classify between absent vs mild and above, for AIMS, we classify between none and minimal vs mild and above, for GAIT, between normal vs everything else. For a more detailed explanation of the considered phenotypes, we refer to the official documentation. The results are reported in Fig. 3. While we cannot conclude that any of the analyzed methods can accurately predict all clinical assessments from structural MRI scans, AnatCL overall achieves the best results three out of ten times, which is more than any other baseline. Interestingly, AnatCL can better predict the overall AIMS score (Fig. 3(b)), hinting that it may be linked to brain anatomy. This link could lead to the discovery of novel biomarkers and should be studied more in depth in future research.

4.3.5. Ablation studies

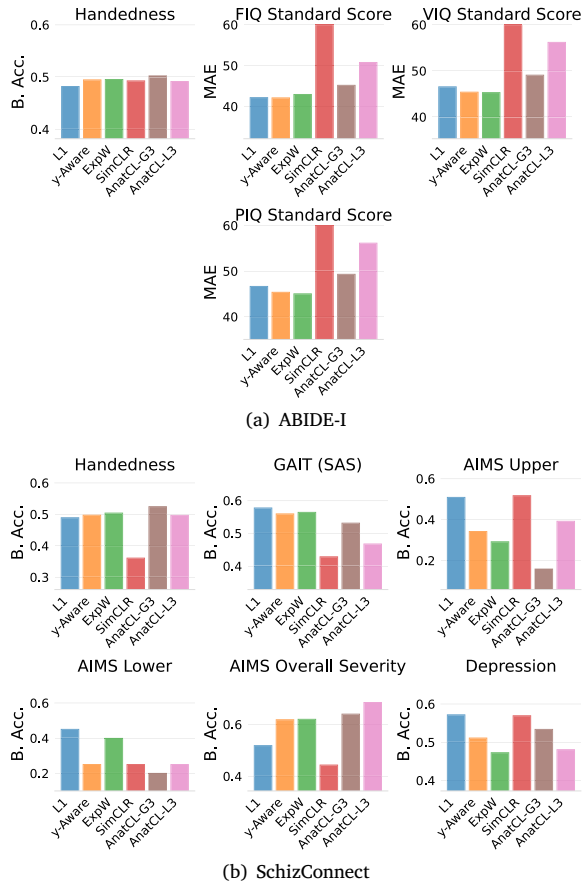
We perform a series of ablation studies focusing on three main components of the proposed framework: (1) the effect of data augmentation, (2) the sensitivity to the weighting hyperparameters λ_1 and λ_2 , and (3) the choice of anatomical features. Additional analyses on model size are provided in the Supplementary Material.

Data augmentation. In the original experiments, no explicit data augmentation is used, as positive pairs are defined based on age and/or anatomy labels. To assess its potential impact, we conduct new experiments comparing AnatCL trained with and without spatial augmentations (random cropping and cutout of $32 \times 32 \times 32$ boxes as for the SimCLR baseline), reported in Table 2 as w/ aug. The results indicate that augmentation does not substantially improve performance, consistent with prior findings on voxel-based morphometry (VBM) data [2,36].

Table 3

Loss function and pre-training strategy ablation. AnatCL learns more generalizable representations by combining anatomical features and age.

Loss function and pre-training strategy ablation. AnafCL leads to more generalizable representations by combining anatomical features and age.																
Method	OpenBHB		Schizconnect				ABIDE-I			ADNI		OASIS-3		Avg.		
	Brain Age (I)	Sex	SCZ (Broad)	SCZ (Strict)	Schizoaffect	Bipolar	Autism	Aspergers	PDD-NOS	HC vs AD	sMCI vs pMCI	HC vs AD				
y-Aware ($\lambda_1 = 0$)	2.66±0.06	79.6±1.13	66.20 ±4.50	71.04±2.31	70.29 ±14.73	63.95±19.81	55.84±3.37	60.60±9.22	56.17±10.17	80.30±1.80	64.72±4.43	76.70±3.30	67.76			
AnatSSL-L3 ($\lambda_2 = 0$)	8.87±0.25	75.64±2.58	57.30±4.36	68.00±10.32	60.41±17.43	55.85±13.46	50.94±3.70	57.56±3.85	45.59±10.85	76.31±6.09	66.79 ±3.33	68.98±8.00	62.12			
AnatSSL-G3 ($\lambda_2 = 0$)	9.12±0.39	72.43±3.14	54.12±7.71	60.76±6.99	54.53±8.71	58.71±14.00	50.40±5.86	50.81±4.29	45.17±6.23	68.12±6.01	63.38±4.00	64.40±6.99	58.44			
AnatCL-L3	2.55 ±0.07	80.49 ±1.25	66.03 ±4.06	69.86±11.20	52.72±6.64	63.12±21.50	53.99±2.60	61.44 ±4.33	56.67 ±7.43	79.71±2.51	64.78±3.57	74.26±3.32	65.73			
AnatCL-G3	2.56 ±0.06	79.0±1.0	64.31±4.43	72.46 ±4.41	62.42 ±14.95	72.42 ±15.77	57.12 ±2.79	58.25±4.46	64.39 ±9.26	80.84 ±2.38	64.88±4.37	78.32 ±4.48	68.58			

**Fig. 3.** Prediction of clinical assessment scores from structural brain MRIs.

Sensitivity to λ hyperparameters. We analyze the influence of the weighting terms λ_1 and λ_2 , which control the balance between age-driven and anatomy-driven representations. Setting $\lambda_1 = 0$ removes the anatomical component, leaving only age as a weak label—this corresponds to the *y-Aware* formulation. Conversely, setting $\lambda_2 = 0$ excludes the age information and retains only anatomical features, yielding a self-supervised variant we denote as *AnatSSL*, since the features can be directly derived from FreeSurfer outputs. We compare all configurations using the Desikan parcellation across all tasks in Table 3. Results show that jointly leveraging both anatomical and age-driven supervision leads to consistently better performance.

Choice of anatomical features. In Table 4, we compare AnatCL using the three anatomical measures adopted in the main experiments with a configuration including all seven available measures. The results show that not all features contribute equally to representation quality, and that the selected subset yields the most stable and accurate embeddings across tasks.

5. Conclusions

We propose AnatCL, a foundation model based on brain anatomy and trained with a weakly contrastive learning approach. With thorough validation on 12 different downstream tasks, we show that incorporating anatomy information during training can result in more accurate predictions of different neurological and psychiatric conditions, and appears promising for clinical assessment scores. We release the weights of the trained model for public use, enabling researchers and practitioners to leverage AnatCL in many different applications.

CRedit authorship contribution statement

Carlo Alberto Barbano: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Matteo Brunello:** Writing – original draft, Software, Methodology, Investigation. **Benoit Dufumier:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization. **Marco Grangetto:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.patrec.2025.11.028>.

Data availability

Data is public. Code and models are provided at <https://github.com/EIDOSLAB/AnatCL>.

Table 4

AnatCL results with all seven anatomical features.

Method	Schizconnect				ABIDE-I			ADNI		OASIS-3 HC vs AD
	SCZ (Broad)	SCZ (Strict)	Schizooff.	Bipolar	Autism	Aspergers	PDD-NOS	HC vs AD	sMCI vs pMCI	
AnatCL-L3 (Desikan)	66.03±4.06	69.86±11.20	52.72±6.64	63.12±21.50	53.99±2.60	61.44±4.33	56.67±7.43	79.71±2.51	64.78±3.57	74.26±3.32
AnatCL-L3 (Destrieux)	67.31±5.75	70.03±7.16	72.47±12.27	55.88±13.23	53.14±4.20	60.99±4.94	56.63±6.74	79.38±3.56	64.97±3.40	77.00±2.28
AnatCL-G3 (Desikan)	64.31±4.43	72.46±4.41	62.42±14.95	72.42±15.77	57.12±2.79	58.25±4.46	64.39±9.26	80.84±2.38	64.88±4.37	78.32±4.48
AnatCL-G3 (Destrieux)	64.12±3.18	73.04±4.42	59.76±15.20	84.19±7.10	55.41±4.55	58.96±7.52	58.10±4.63	80.75±3.35	65.91±2.23	78.66±2.89
AnatCL-L7 (Desikan)	66.16±6.12	70.92±8.82	63.65±18.60	66.81±20.04	55.28±1.41	60.76±7.85	59.42±8.02	80.17±1.98	64.59±1.75	74.22±4.21
AnatCL-L7 (Destrieux)	68.06±4.93	70.51±10.15	63.73±17.27	60.56±14.39	55.79±2.25	59.68±1.02	56.28±3.90	81.10±2.21	63.92±5.43	77.33±4.91
AnatCL-G7 (Desikan)	68.16±5.95	74.06±8.34	61.97±15.20	69.33±19.83	55.60±2.37	58.04±5.92	57.15±6.08	80.79±1.32	63.46±3.46	75.69±3.22
AnatCL-G7 (Destrieux)	66.98±3.32	72.39±6.55	59.66±16.00	67.04±14.83	54.47±3.96	60.85±5.79	55.99±6.90	79.08±2.85	64.45±2.13	75.93±3.81

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